Cover page

Built Environment Case Competition 2022

Pillar: Productivity

Smart Construction System for DfMA

(SCS-DfMA)

Team WXYZ

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1. Problem Background

Productivity of the traditional construction industry is declining throughout the years [1]. Therefore, Singapore implemented Design for Manufacturing & Assembly (DfMA) as it shows high potential in improving productivity and reducing labour cost [2], e.g., multiple projects using prefabricated prefinished volumetric construction (PPVC) [3]. However, DfMA faced hindrances, affecting its adoption in the construction industry (Fig.1).

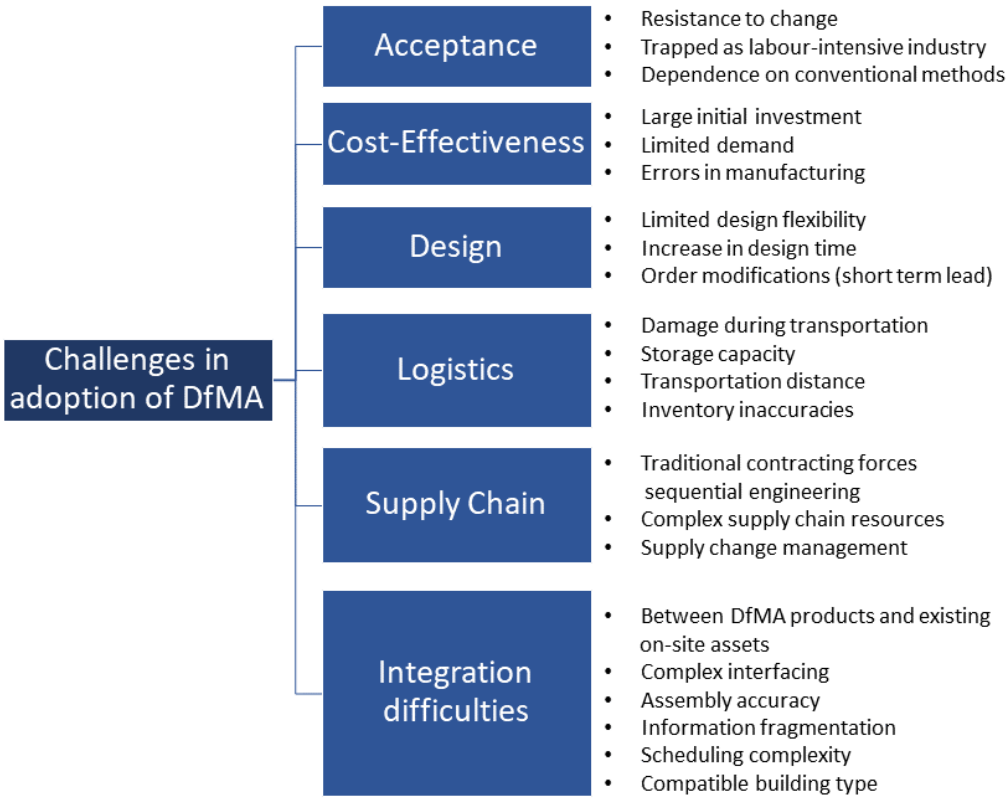


Fig. 1. Challenges in the adoption of DfMA [4,5]

Quality control is vital in DfMA, especially when it is done overseas. Critical factors in off-site manufacturing include material quality, equipment operation error, component accuracy deviation, and manufacturing capacity [4]. Moreover, damage during transportation and inventory inaccuracies cause double handling and waste [5].

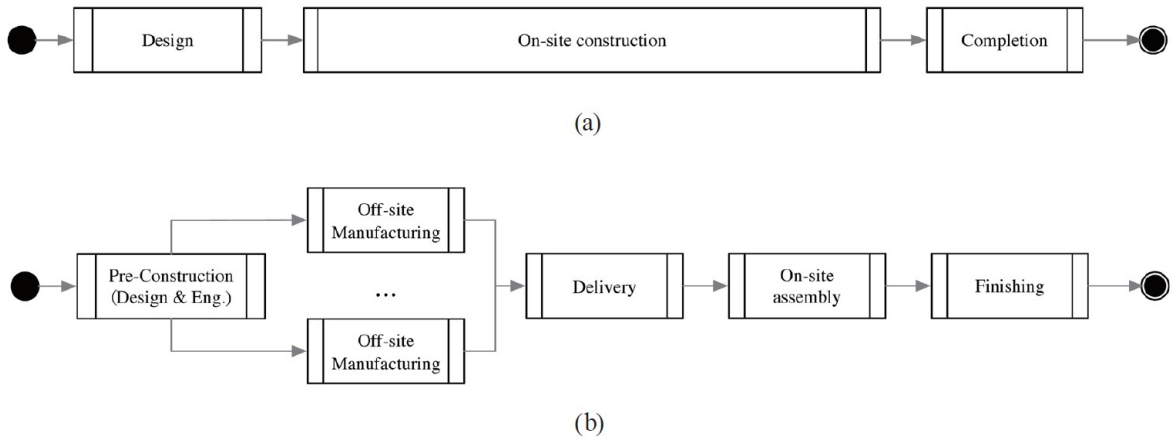


Fig. 2. Comparison of the processes of traditional construction and PB construction. (a) Traditional construction process. (b) PB construction process. [4]

As shown in Fig.2., DfMA construction process consists of more complex processes and relationships. Therefore, integrating suitable technologies, e.g., Internet of Things (IoT) in DfMA is required to increase its feasibility in the industry.

2. Solution

2.1 Background

To counter the low degree of informatization in the built environment, our team proposes a conceptual framework of an integrated digitalization approach, through implementing IoT into DfMA, named Smart Construction System for DfMA (SCS-DfMA), which aims to improve the efficiency of communication and collaboration between government supervision departments, contractors, consultants, designers, and clients.

IoT combines acquisitions, communication, and computing technologies. It facilitates information-sharing between human-machine-object, utilizing data collected by sensors and smart devices for automation. With IoT, information is shared at all phases of design, component prefabrication, transportation, and assembly, reducing timeline, hence increasing efficiency. (Table 1)

Table 1. Study on implementation of IoT in construction [6]

Phase	KPI	Before	After	Improvement
Production	Paper work	200–300 A4 papers/day	110–145 A4 papers/day	48.3% averagely
	WIP inventory	110 sets in average	98 sets in average	10.9%
	Production lifecycle	10 days in average (façade)	6 days in average (façade)	40%
	Time cost on locating	7–8 min in average	5.5 min in average	31.25%
Logistics	Paper work	5 papers/car	3 papers/car	40%
	Waiting time for delivery	2 h	1.5 h	25%
	On time rate of delivery	92.5%	99.8%	7.3%
	Order picking time	2 h/car	1.2 h/car	40%
On-site assembly	Assembly time	6 day per cycle	5.6 day per cycle	6.67%

2.2 Development of the SCS-DfMA framework

The proposed conceptual framework of SCS-DfMA comprises of four systems, namely PIIC, IBS, GMTS and DISEP, as shown in Fig. 3 and 4 [7].

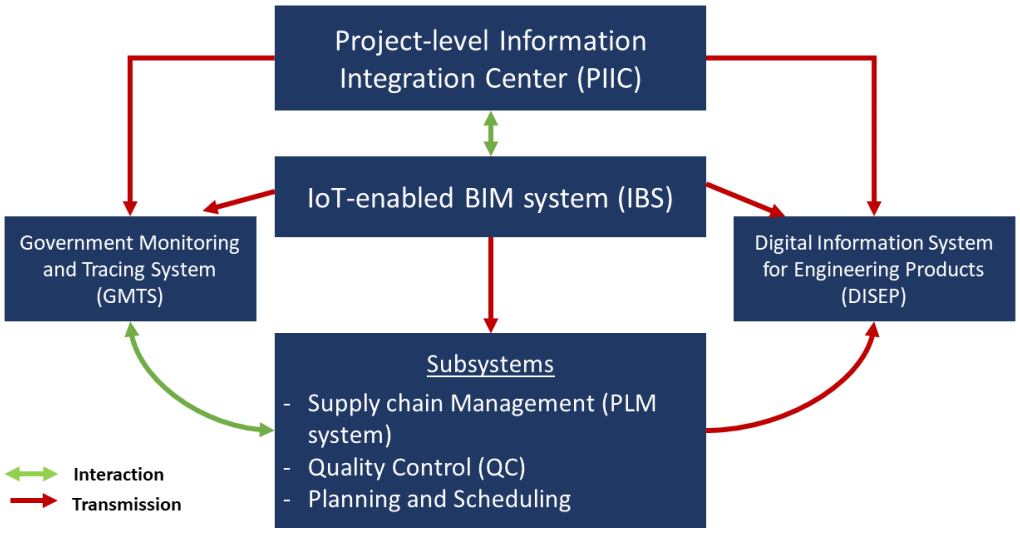


Fig. 3. Proposed SCS-DfMA Conceptual Framework

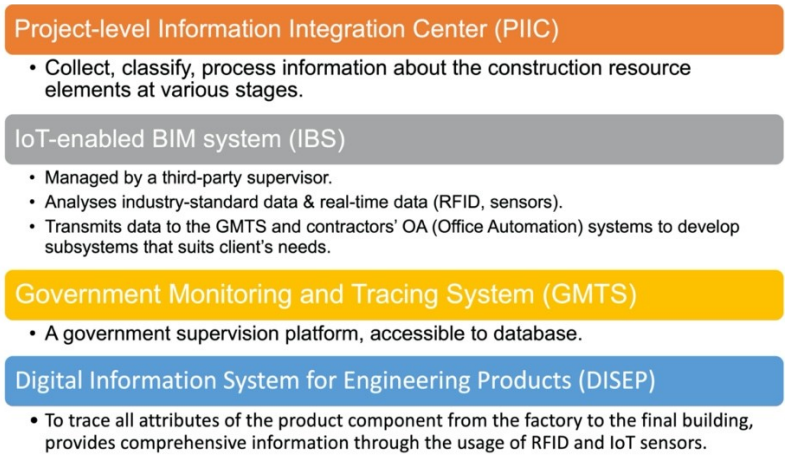


Fig. 4. Functions of the SCS-DfMA systems [7]

2.3 Operating mechanism of the SCS-DfMA

i. Pre-construction & Design Phase:

As shown in Fig. 5, the information interaction between the whole construction project design, manufacturing and construction processes is realized under the implementation of IoT, which ensures the integrated construction approach of design-manufacturing-assembly.

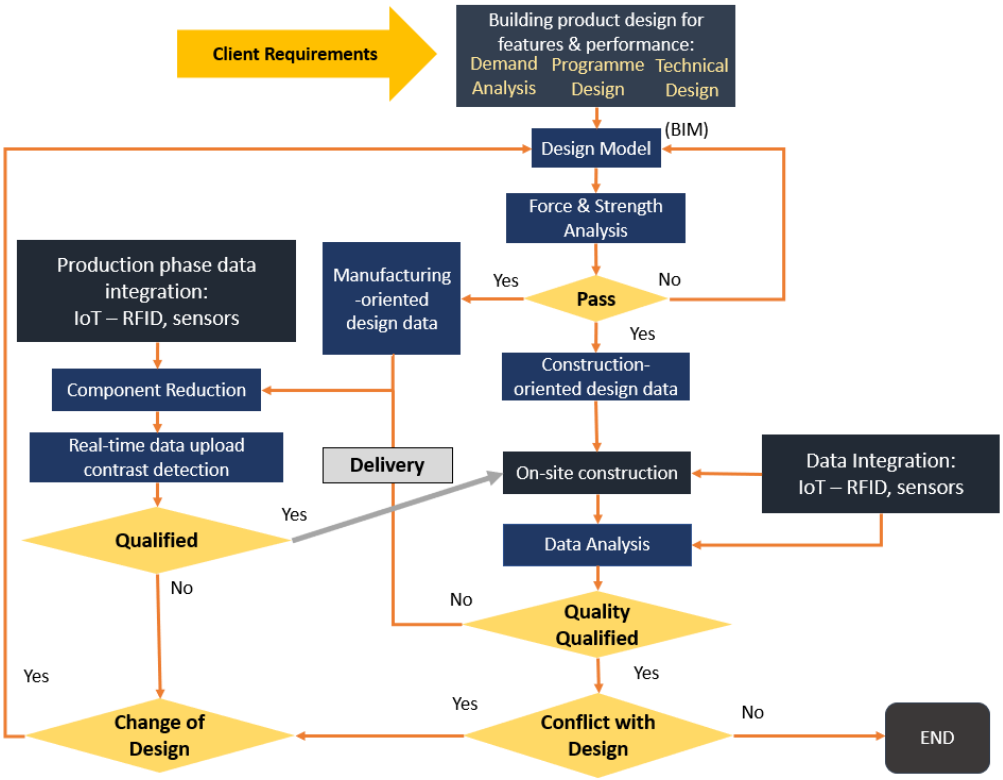


Fig. 5. Operating mechanism of SCS-DfMA in the design phase [7]

ii. Component Production (Off-site Manufacturing) Phase:

During the component production phase of SCS-DfMA, data that has been collected through RFID and sensors undergoes information processing and decision-making analysis, as shown in Fig. 6.

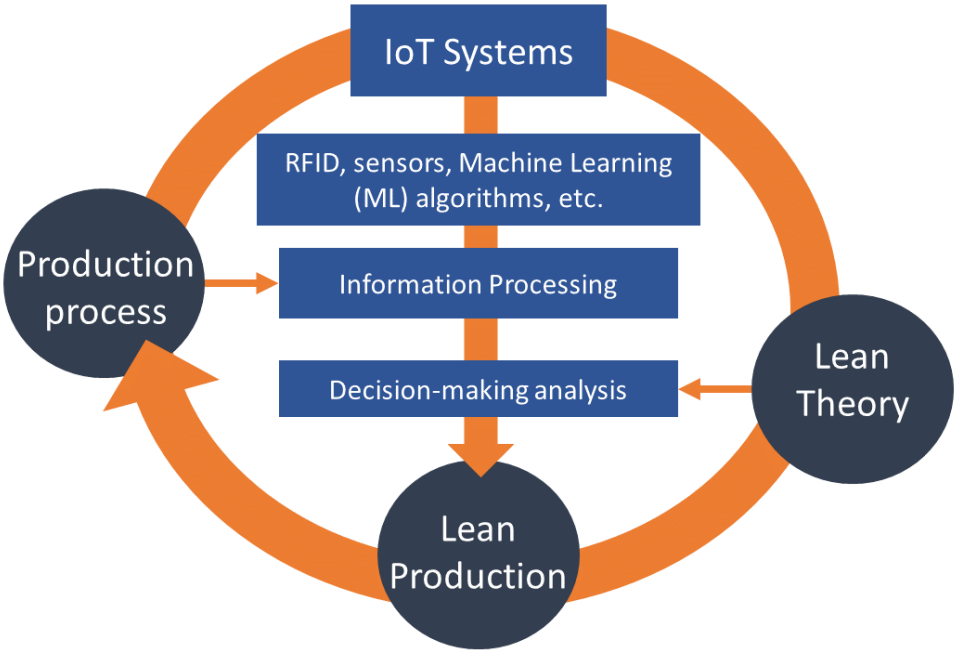


Fig. 6. Operating mechanism of SCS-DfMA in the component production phase [7]

iii. Product Delivery (Transportation & On-site Assembly) Phase:

In the product delivery phase, tracking and optimization of transport routes can be done through the usage of RFID, GPS, and GIS. The DISEP* system collects and integrates actual engineering data with design data to produce digital engineering products (Fig. 7). On-site management can be carried out by supervisors through various subsystems (e.g., Project Quality Traceability and Quality Control subsystem) for better control of resources and reduce the occurrence of safety accidents.

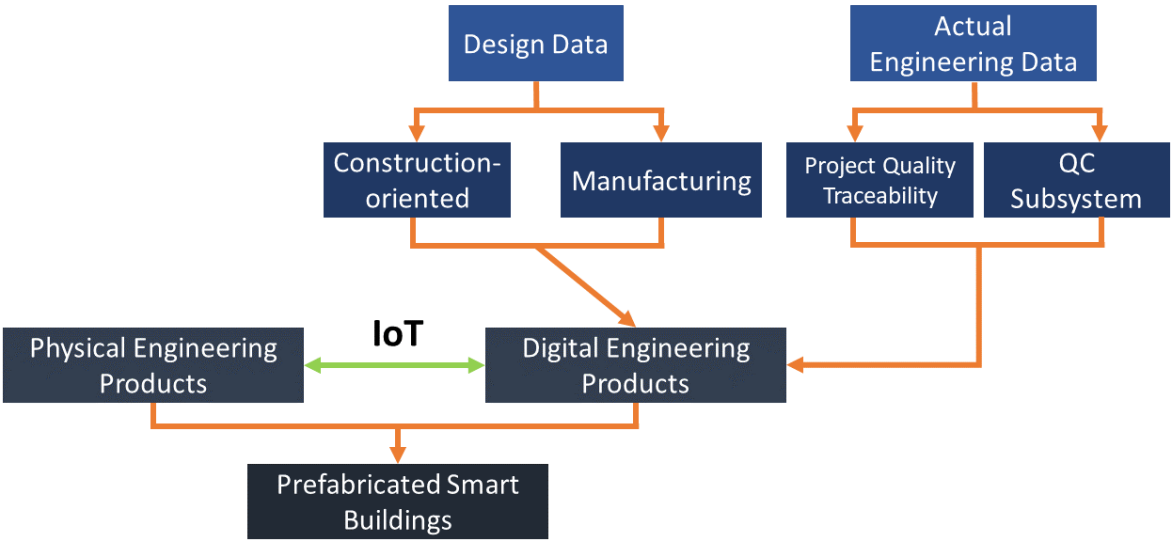


Fig. 7. Operating mechanism of SCS-DfMA in the product delivery phase [7]

DISEP: Digital Information System for Engineering Products

2.4 The implementation of SCS-DfMA & IBS on concrete production

The proposed framework will be utilizing off-site concrete module manufacturing as an example to showcase the efficiency and performance of SCS-DfMA in the process of productivity improvements in the construction industry.

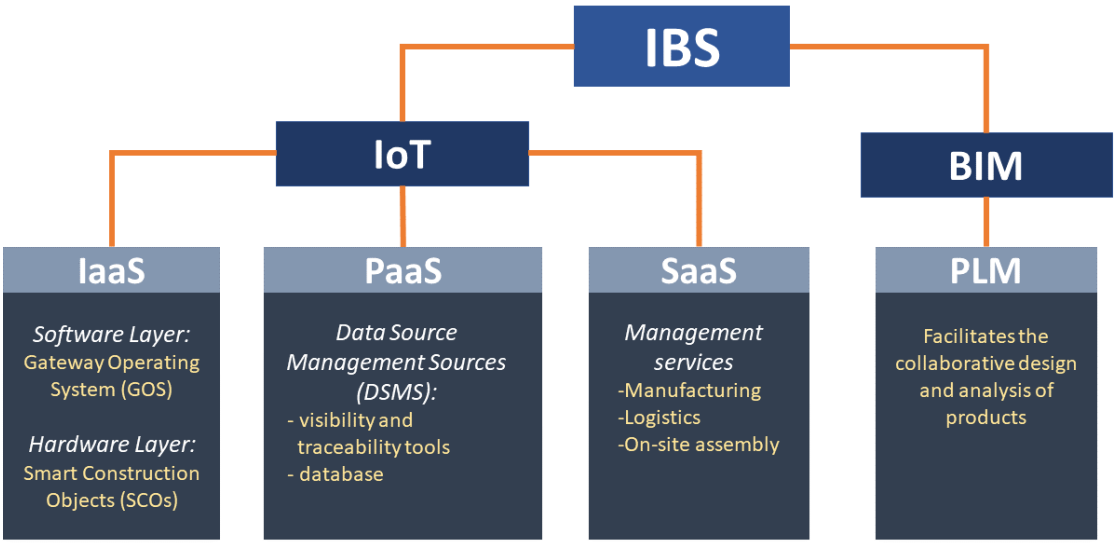


Fig. 8. Overview of multi-dimensional IoT-enabled BIM System (IBS) [8]

Based on Auto-ID and Information technologies, IBS integrates information such as project cost to extend the conventional BIM system, which could only provide 3D models, to a multi-dimensional platform. To capture, integrate, and share product-based information (e.g., concrete) generated by BIM-based analysis, Product Lifecycle Management (PLM) system is implemented, and its function is as shown in Fig. 9 [10].

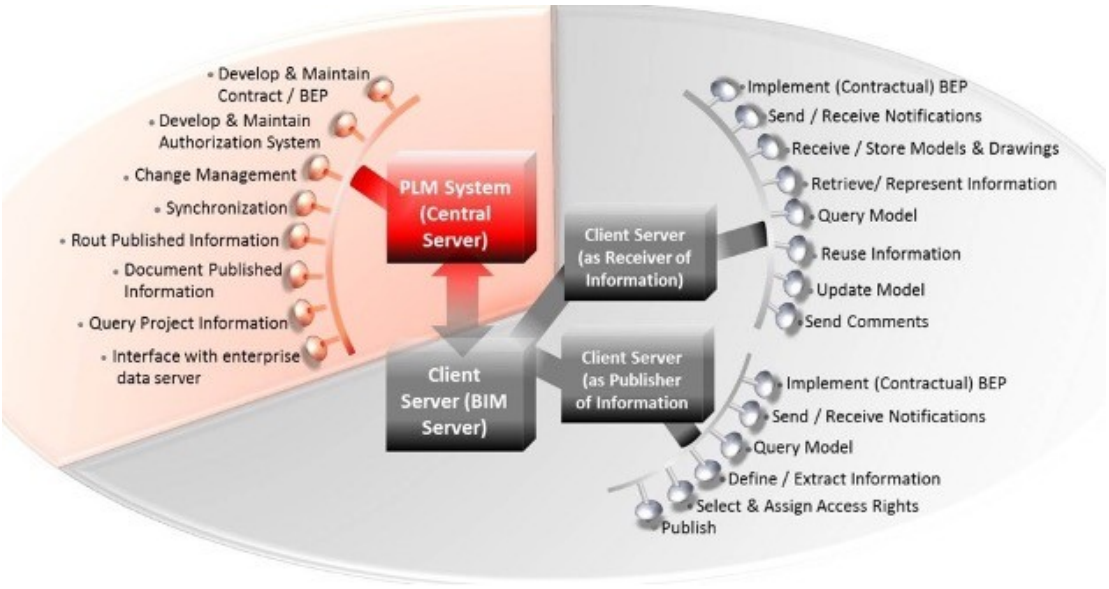


Fig. 9. Integration of BIM and PLM systems [10]

The IoT system under IBS* is hosted on the cloud over a network that comprises of 3 levels, namely infrastructure as a service (IaaS), platform as a service (PaaS) and software as a service (SaaS) (Fig. 8) [8]. In IaaS, Gateway Operation System (GOS) connects with Smart Construction Objects (SCOs), e.g., concrete modules that are attached with RFID devices and sensors, providing real-time information in a standardized format to upper-level decision-making systems (Fig. 10) [8]. Complex event processing (CEP) technology is used to integrate information into a standardized scheme [8].

**IBS: IoT-enabled BIM System*

Referring to a case study by Scott Fowler, a simple Singular Value Decomposition (SVD) Machine Learning multilinear regression model can be applied in IBS as a part of the decision-making system, where the optimal environment for drying concrete can be determined when the optimal predicted data is obtained based on actual measurements [9].

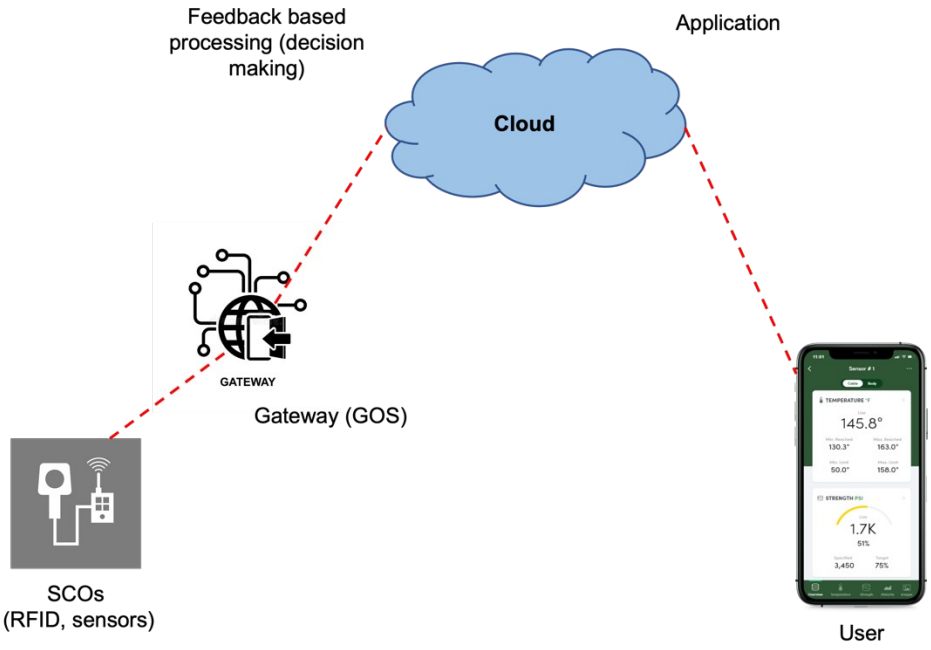


Fig. 10. Overview of the cloud system

In IBS*, sensors will be used for lifecycle concrete monitoring which the application examples are shown in Table 2 [11,12]. All sensors will be connected through LPWPAN (Low-Power Wide-Area Network) to collect and transmit data to the project quality traceability subsystem through the Gateway. It is connected to the GMTS* for authorities to promptly correct any misconduct behaviors that do not meet the standard, as real-time data enables parties to monitor project status. Traceability tools can further enhance data sharing and interoperability between BIM and its subsystems [8]. By doing so, the quality of the concrete produced can be highly assured during off-site manufacturing before the delivery phase takes place.

**IBS: IoT-enabled BIM System*
**GMTS: Government Monitoring and Tracing System*

Table 2. Application of IoT sensors [11,12]

Temperature and humidity sensor	Placed on the surface of the concrete, and a temperature sensor is attached to rebars inside of the concrete structure to monitor temperature difference between surface and core temperature of the concrete element.
Viscosity sensor	Monitor the real time workability of concrete to ensure optimization of vibration imparted.
Piezoelectric (PZT) sensor	Sensor signal measured by a PZT sensor is decomposed into sub-signals by the wavelet packet algorithm. Damage index (difference in the energy vector of healthy and damaged state) provides real-time information regarding occurrence and propagation of cracks.
3D laser	Quality inspection of concrete based on comparison of as-built models obtained from laser scanning with the corresponding design CAD models. The damage detection technique can be based on the computation of the mean and Gaussian curvatures of the concrete surface as well.

3. Impacts

3.1 Business benefits

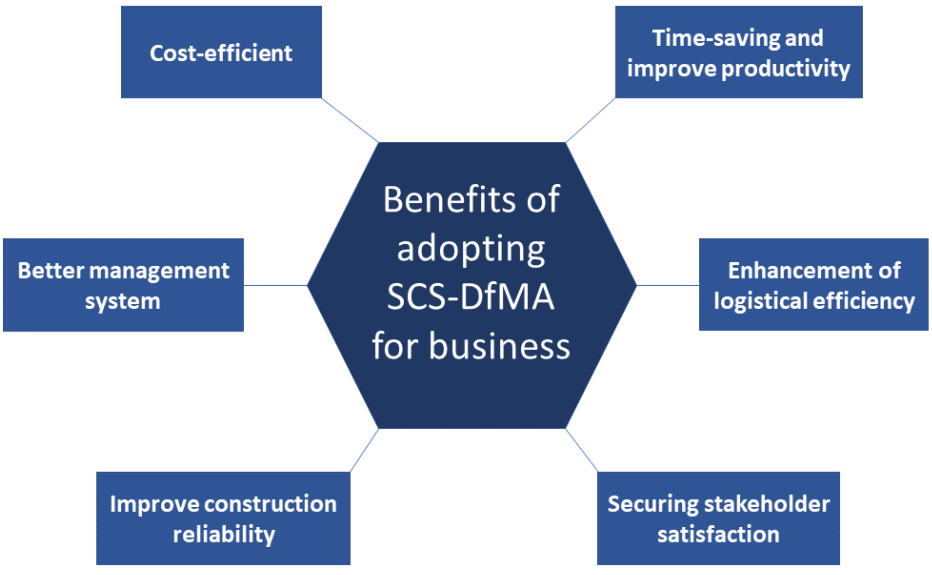


Fig. 11. Benefits of adopting SCS-DfMA for business [5,15]

SCS-DfMA creates a closed-loop visibility and traceability mode by enabling various end-users to monitor a project's progress and cumulative cost in real time. Therefore, stakeholders can concentrate on different areas and handle unexpected deviations throughout the life cycle of a prefabrication-based building project. [6]

By raising awareness of the benefits and efficiency of SCS-DfMA, as illustrated in Fig. 11, newly evolved technology such as SCS-DfMA will be more widely accepted by stakeholders. Additionally, the progress of the construction project can be shortened, as seen in Fig. 12. Hence, the adoption of SCS-DfMA will increase the productivity and quality performance of building operations. [16]

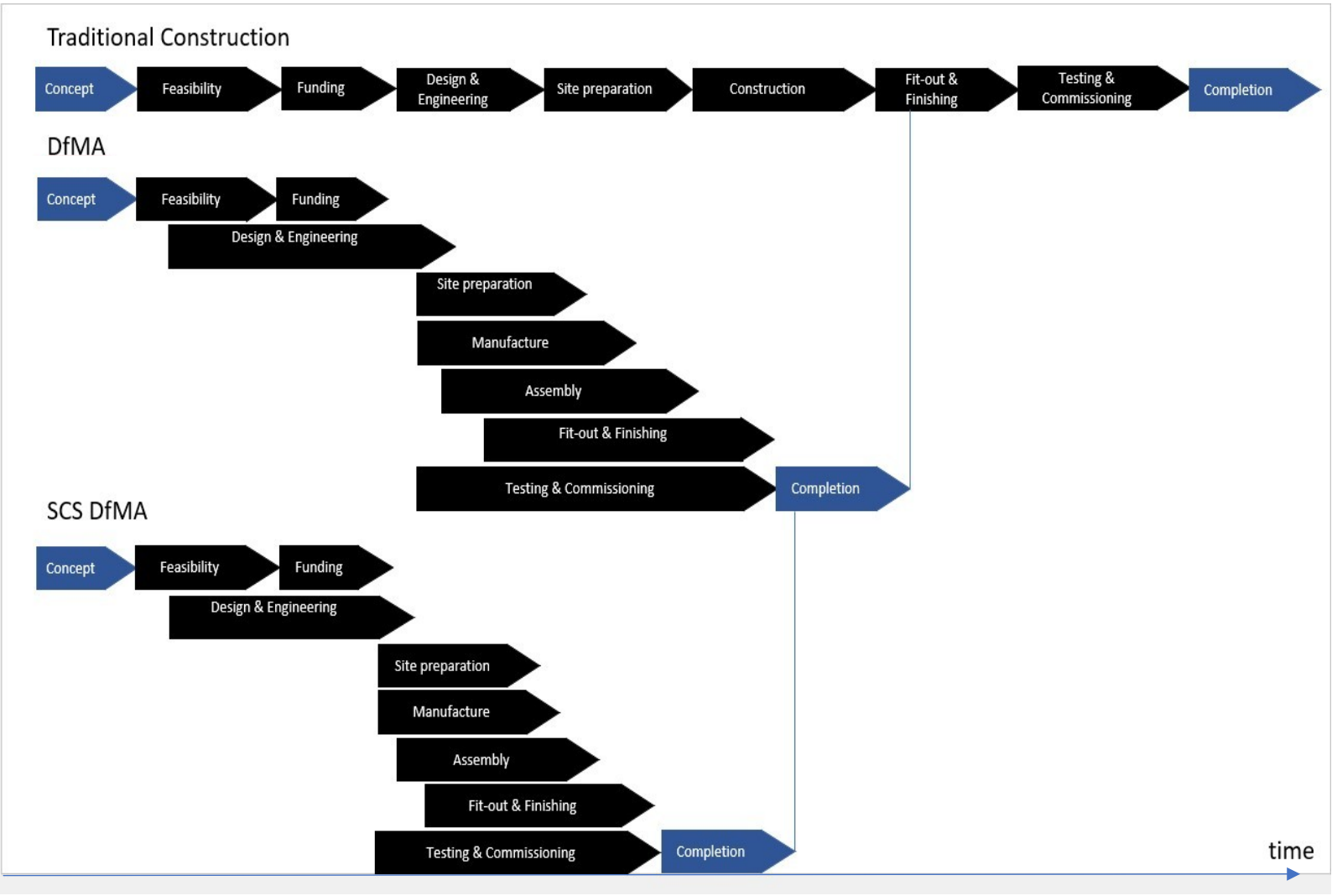


Fig. 12. Comparison between the timelines of traditional construction, DfMA, and SCS-DfMA methods [17]

3.2 Environmental impacts

As illustrated in Fig.13. , SCS-DfMA reduces carbon footprint by reducing waste, conserving electricity, and other aspects such as less paperwork and automatically turning off idle equipment.

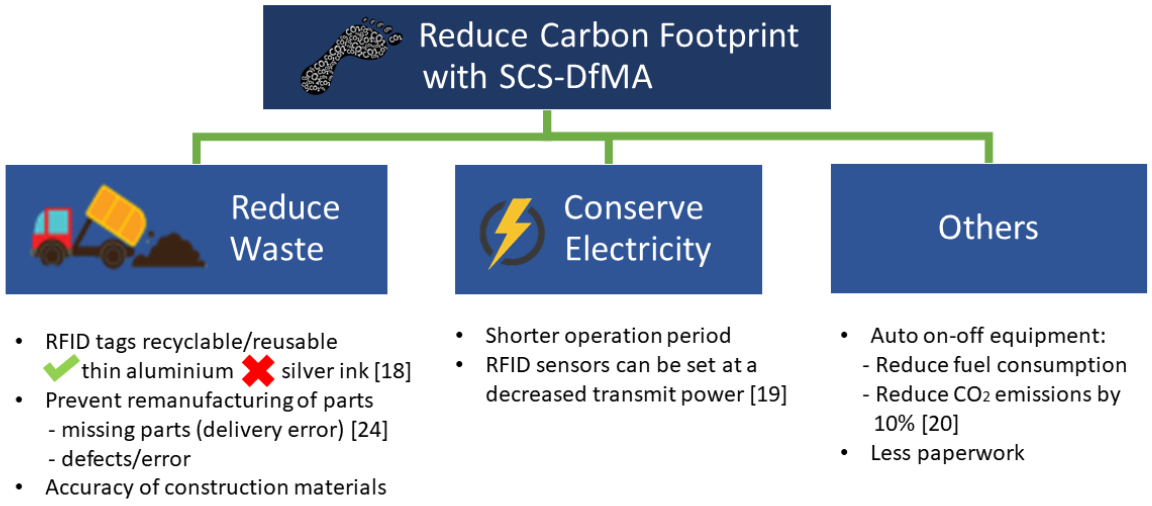


Fig. 13. Reduce carbon footprint with SCS-DfMA

3.3 Economic analysis

Items	DfMA (SGD)	SCS-DfMA (SGD)
Labour Cost ***Assume SCS-DfMA require less manpower and time.**	8 workers + 2 site managers 59 weeks x 44 hrs x SGD16.5 x 10 **428,340**	8 workers + 1 site managers 50 weeks x 44 hrs x SGD16.5 x 9 **326,700**
Error Budget - Double handling - Delivery issues	**10,000**	**5,000**
RFID Tags ***Assume 1 tag per module**	-	824 x 0.15 **=123.6**
Sensors -RFID sensors -Thermal sensors -Infrared sensors	-	100,000/10 **=10,000**
Software ***estimate 10,000/yr**	-	(10,000 x 1 year)/10 **=1,000**
Total	**438, 340**	**342,823.6**

Case Study: Modular Construction in Wolverhampton [21]
- 3 blocks of 8 to 25 stories
- 824 modules
- Floor area = 20730 m^2
- Total duration = 59 weeks

Average wage of worker = SGD 16.5/ hr [22]

Assume cost distributed across 10 projects

21.79% saved

Fig. 14. Comparison of costs between SCS-DfMA and DfMA

As shown by the rough estimation in Fig. 14., SCS-DfMA can be more cost efficient than regular DfMA. Hence, it is possible to execute projects with similar costs with increased productivity.

The cost of implementing SCS-DfMA includes equipment costs which approximates to $4000 per sensor including cabling, RFID tags (US$0.07 - US$0.15) [23], installation, software, licensing, and maintenance.

However, the implementation of RFID and IoT also reduces errors [24] and labour costs. It successfully reduces costs by the reduction of:

- Waste (Refer 3.2)
- Labour costs due to reduced activity duration [24] (up to 40% increase in productivity [25])

Furthermore, the implementation of IoT is a great way to avoid double-handling, which is a major contributor to construction waste [26].

4. Challenges of SCS-DfMA

A summary is shown in Fig.15.

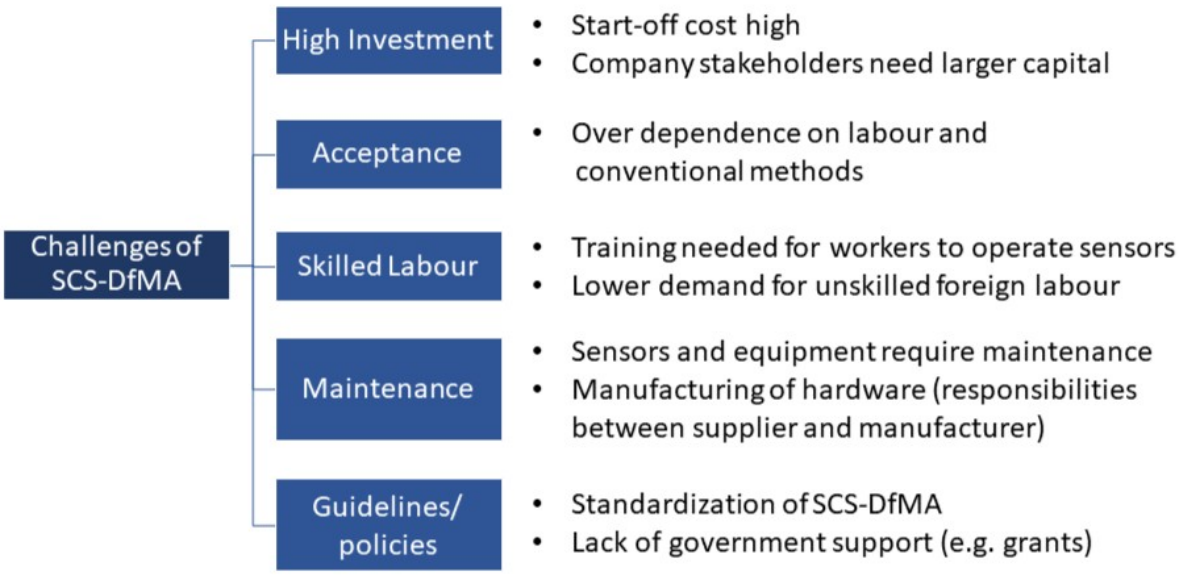


Fig.15. Challenges of SCS-DfMA

Although SCS-DfMA shows many beneficial impacts to the industry, the reluctance to change due to over dependence on traditional methods remains a hurdle. Cooperation between all parties is needed to progress effectively. Along with the demand for skilled labours, professional training is required to support the workflow. This means lesser jobs for unskilled labour.

SCS-DfMA requires initial high investment and long return years. Thus, support (e.g., grants) from the government is crucial to push SCS-DfMA into the market as a conventional method.

5. Conclusion

Our proposed solution, SCS-DfMA, implementing IoT with the concept of DfMA into the construction will greatly increase the productivity as DfMA method relies heavily on logistics, quality control, and design planning. With real-time monitoring and data, we can optimize decision-making in all aspects, achieving a more feasible and effective DfMA.

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Appendix

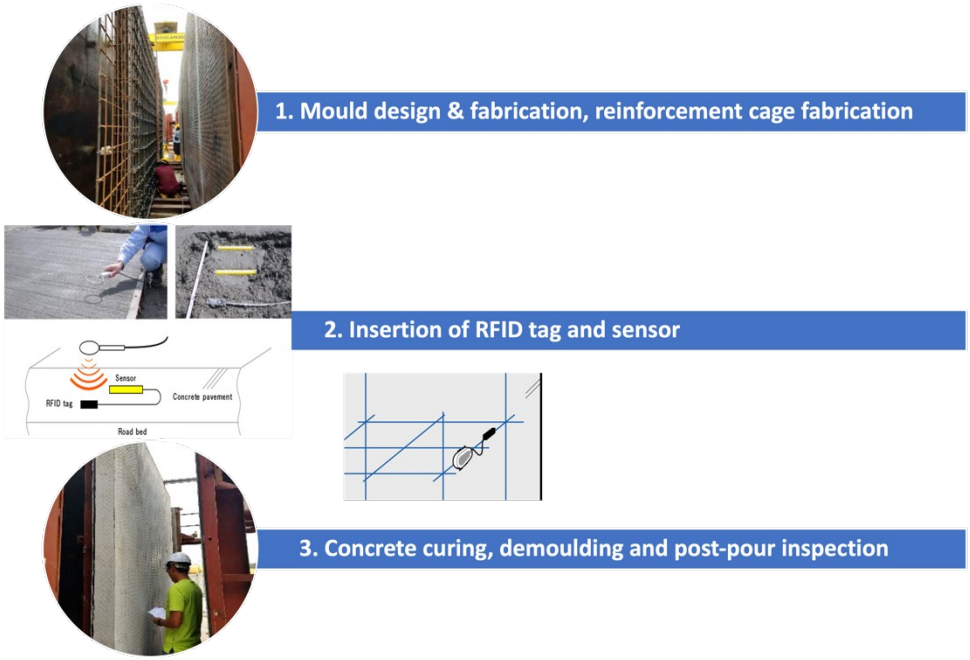


Fig. 16. Off-site reinforced concrete module production process with real-time data collection [3,13]

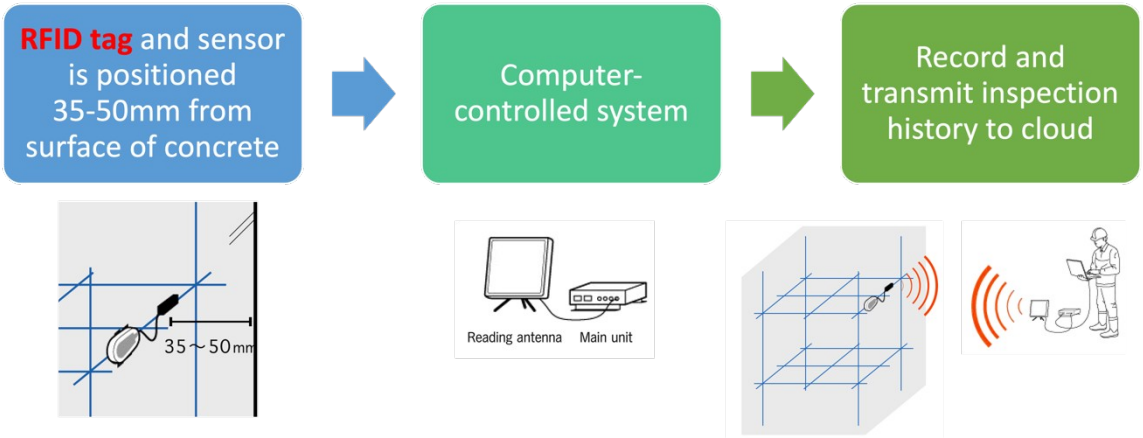


Fig. 17. Real-time data monitoring using RFID & sensors [13]

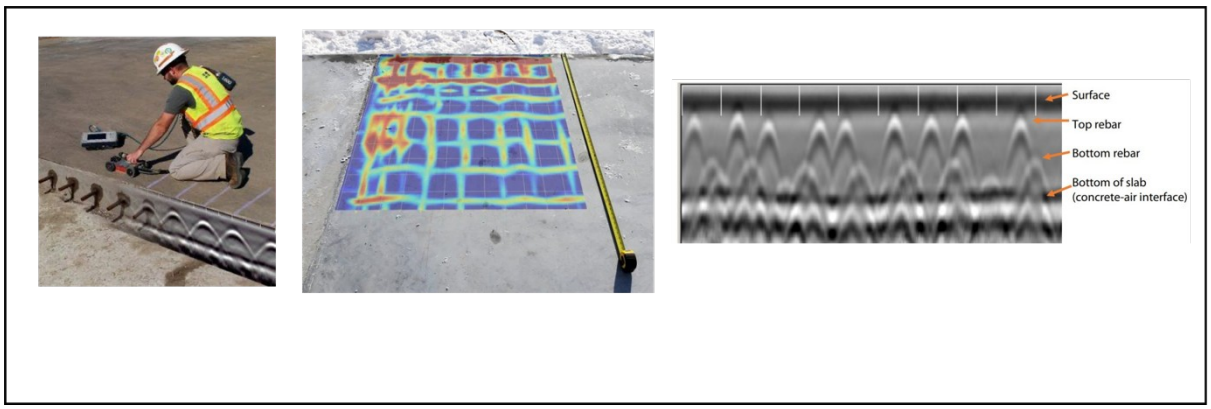


Fig. 18. Concrete quality inspection using Ground Penetrating Radar (GPR) [14]

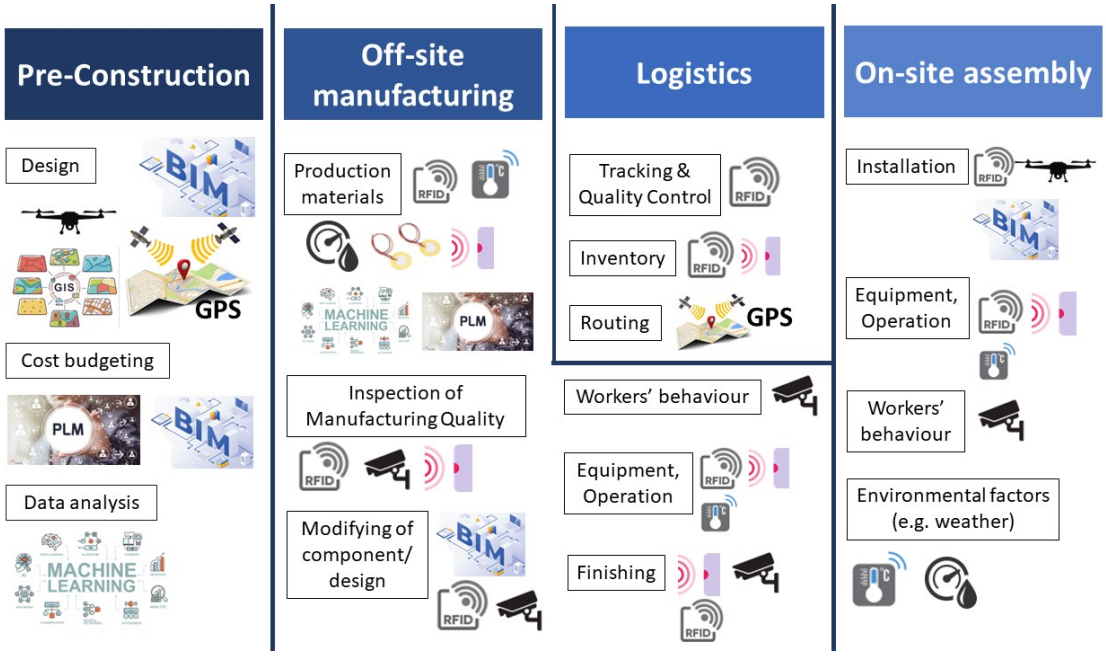


Fig. 19. Overall application of SCS-DfMA in 4 phases of construction